Kaluza's fifth dimension in a Kron machine

Is electric charge motion along a fifth dimension, and is Kron's machine the same construction?
CLAUDE.md §3a and §3f

kronspice simulation report · 2026-11-07 · c = G = 1 (Gaussian) for the geodesic; declared machine values

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Summary

Th. Kaluza (1921) added a fifth dimension to Einstein's space-time and found that one five-dimensional geometry holds both gravity and electromagnetism: a charge moving freely in 5-D is seen in 4-D as gravity plus the Lorentz force, and electric charge is momentum along the fifth dimension. Kaluza assumes nothing depends on the fifth coordinate (his "cylinder condition"); Hoffmann (1949) uses the same words for Kron's machines, whose winding charges never appear in the inductances. This example checks Kaluza's equations, solves the exact 5-D geodesic of a charge in a magnetic field, reproduces his difficulty with the electron and Klein's answer, and builds the same construction as a two-coil Kron machine solved three ways.

- **Kaluza's equations hold as printed.** His closed forms (eq. 3) agree with the three-index symbols of his metric to 6.8×10^{-27} with his α and 1.8×10^{-13} with α = 0.7. His coupling α = √(κ/2) = 3.0548×10^{-14} (he printed 3.06×10^{-14}).
- **Charge is fifth-dimension motion.** The exact geodesic is a Lorentz-force orbit with $q/m = k\cdot p_w/\sqrt{(1 + p_w^2)}$, $k = 2\sqrt{G}$; orbit radius and frequency match the Lorentz force to all printed figures for p_w from 0.1 to 10.
- **A ceiling on charge.** $q/(m\sqrt{G})$ can never exceed 2; the electron needs 2.041×10^{21} and the proton 1.112×10^{18}, so no real particle fits on a classical 5-D geodesic (Kaluza's own "serious difficulty", p. 971).
- **Klein's circle** has radius $R = 2\sqrt{G}\cdot\hbar/(c\cdot e) = 2\, l_P/\sqrt{\alpha} = 3.784\times10^{-34}$ m; its first charged mode weighs 5.21×10^{17} GeV, near the Planck mass, not an electron.
- **The Kron machine is the same construction.** Kron's geodesic in (q1, q2, θ), the rotor alone with the flux linkages held, and the kronspice machine element agree: 1.1×10^{-10} rad/s and 3.7×10^{-4} rad/s largest speed differences; kronspice ledger mismatch 3×10^{-11}.
- **No energy from the extra dimension.** The 5-D geodesic keeps its norm and the machine its energy (energy drift 6×10^{-15}).

Background

Kaluza's metric (his eq. 2) frames Einstein's 4-D metric with the electromagnetic potential. The modern form used for the geodesic is $\hat{g} = (\eta + k^2AA, kA; kA, 1)$, with $k = 2\sqrt{G}$, η the flat 4-D metric, A the vector potential and w the fifth coordinate. Because nothing depends on w, the momentum p_w along it is conserved; the 4-D projection of the free-fall path is then a charged particle's orbit. Kaluza's eq. 8a, $\rho0/\mu0 = 2\alpha\cdot u^0$, is the small-momentum part of this relation.

In a Kron machine the coordinates are the winding charges and the rotor angle θ, and the inductances (the metric) depend on θ only. The charges are therefore cyclic coordinates, and their momenta, the flux linkages, are conserved, exactly as p_w is. Eliminating them (Routh reduction) leaves the rotor alone in a potential set by the conserved fluxes: the machine's version of the force the fifth dimension leaves behind. This is not a hypothesis-register entry; it is a check of a historical theory and of the structure of Kron's machine equations.

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Results

1. Kaluza's equations

quantity	computed	Kaluza / today
$\alpha = \sqrt{(\kappa/2)}$, $\kappa = 8\pi G/c^2$	3.0548×10^{-14}	3.06×10^{-14} (printed)
eq. 3 closed forms vs symbols of eq. 1 metric (his α)	largest difference 6.8×10^{-27}	holds as printed
eq. 3, with α = 0.7	largest difference 1.8×10^{-13}	holds as printed
electron e/m	1.77×10^{7} is e/m in electromagnetic units per gram	today 1.7588×10^{7}

2. A charge in a magnetic field as free motion in five dimensions

Exact 5-D geodesic, c = G = 1, Gaussian charge, uniform B = 0.05 along z, speed 0.3 c.

p_w	q/m	orbit radius	Lorentz radius	ω	Lorentz ω	p_w drift
0.10	0.1990	31.60540	31.60540	0.009492	0.009492	6.7×10^{-12}
0.30	0.5747	10.94442	10.94442	0.027411	0.027411	2.7×10^{-13}
1.00	1.4142	4.44750	4.44750	0.067454	0.067454	1.7×10^{-14}
3.00	1.8974	3.31497	3.31497	0.090499	0.090499	9.6×10^{-15}
10.00	1.9901	3.16054	3.16054	0.094920	0.094920	6.8×10^{-15}

3. Kaluza's difficulty with the electron, and Klein's answer

quantity	value
ceiling on $q/(m\sqrt{G})$ for any 5-D geodesic	k = 2
electron $e/(m\sqrt{G})$	2.041×10^{21}
proton $e/(m\sqrt{G})$	1.112×10^{18}
Klein radius $R = 2\sqrt{G}\cdot\hbar/(c\cdot e) = 2\, l_P/\sqrt{\alpha}$	3.784×10^{-34} m (about 23 Planck lengths)
first charged mode $\hbar/(Rc)$	5.21×10^{17} GeV

Klein (1926) makes the fifth dimension a tiny circle, so quantum momentum around it, $p_w = n\hbar/R$, makes charge come in whole steps n·e. The files record a correction: an earlier written review gave 1.3×10^{-33} m, which paired the Heaviside coupling $\sqrt{(16\pi G)}$ with the Gaussian charge, a factor $\sqrt{(4\pi)}$ too large.

4. The same construction in a Kron machine

A stator coil and a rotor coil with mutual inductance M·cos θ and lossless, shorted (superconducting) windings. Declared values: L1 = L2 = 1.0 H, M = 0.5 H, J = 0.01 kg·m²; start i1 = 1.0 A, i2 = 0.5 A, θ = 0.3 rad, at rest. Conserved flux linkages ψ1 = 1.2388 Wb, ψ2 = 0.9777 Wb (Kron geodesic: drift 4×10^{-15}; energy drift 6×10^{-15}).

t (s)	Kron's geodesic in (q1, q2, θ)	rotor alone, fluxes held	kronspice machine element
0.1	−0.707296 rad/s	−0.707296 rad/s	−0.706973 rad/s
0.5	−0.923503 rad/s	−0.923503 rad/s	−0.923217 rad/s
1.0	1.439010 rad/s	1.439010 rad/s	1.438928 rad/s
1.5	−1.340890 rad/s	−1.340890 rad/s	−1.341042 rad/s
2.0	0.666266 rad/s	0.666266 rad/s	0.666265 rad/s

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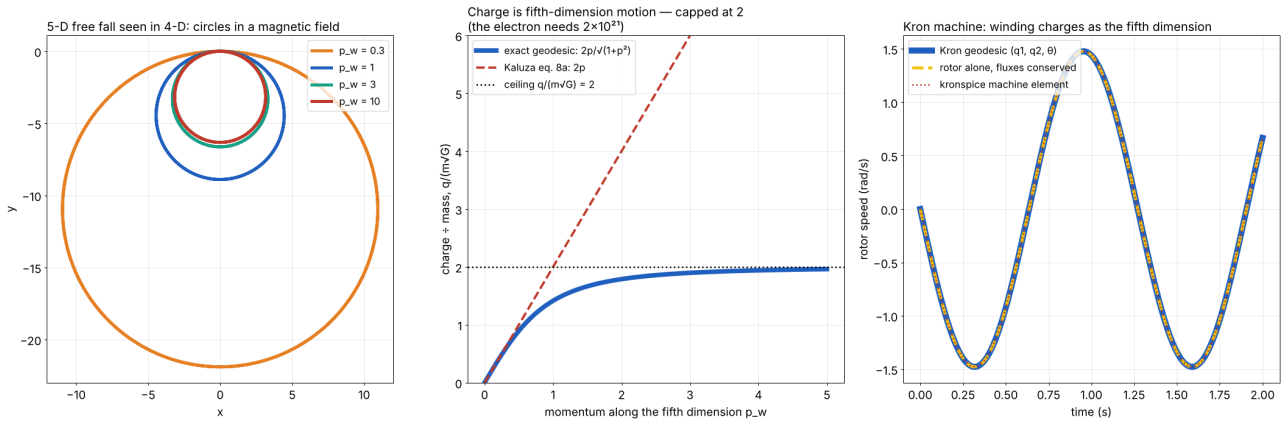


Figure 1. Left: 5-D free fall seen in 4-D, circular orbits in the magnetic field for p_w = 0.3, 1, 3 and 10. Centre: charge-to-mass ratio against momentum along the fifth dimension; the exact geodesic $2p/\sqrt{(1 + p^2)}$ (blue) saturates at the ceiling 2 (dotted), while Kaluza's small-momentum eq. 8a, 2p (dashed red), does not. Right: the Kron machine's rotor speed solved three ways, overlapping.

- The rotor-alone solution matches Kron's geodesic to 1.1×10^{-10} rad/s, and kronspice to 3.7×10^{-4} rad/s (its windings are shorted through 10^{-7} Ω).
- Because the charges never appear in the inductances (the cylinder condition), their momenta, the flux linkages, are conserved, like p_w. Kron's machine is a working Kaluza model.

Ledger and validation

- Momentum p_w conserved to between 6.8×10^{-15} and 6.7×10^{-12} over the geodesic runs.
- Machine: flux linkages conserved to 4×10^{-15}, energy to 6×10^{-15}; kronspice ledger mismatch 3×10^{-11}.
- Tests (*main_test.go*): eq. 3, geodesic = Lorentz force, Klein numbers, machine solved three ways.

Limits and where it stands

- The geodesic uses the modern form of Kaluza's metric with the corner fixed (no scalar field).
- The Kaluza–Klein mathematics lives on as a standard tool in string theory and supergravity.
- No extra dimension has been observed: gravity follows Newton's inverse-square law down to 52 μm (Lee et al., Phys. Rev. Lett. 124, 101101, 2020), and a circle of Klein's size is far beyond any experiment.
- For kronspice the result is structural: Kaluza's geometry is exactly the structure of Kron's machine equations, and the machine shows it working. No energy comes from the extra dimension.

Reproducing the results

```
go run ./examples/kaluza_kron -viz examples/kaluza_kron/viz   # about 1 second
python3 examples/kaluza_kron/viz/render.py
go test ./examples/kaluza_kron
```

Sources: Th. Kaluza, Sitzungsber. Preuss. Akad. Wiss. Berlin (Math. Phys.) 1921, 966–972 (V. T. Toth's translation); O. Klein, Z. Phys. 37, 895 (1926); B. Hoffmann, Rev. Mod. Phys. 21, 535 (1949); J. G. Lee et al., Phys. Rev. Lett. 124, 101101 (2020).

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